

Honors Analysis

Second midterm

November 19th, 2025

Question 1. Write explicit examples of the following. No justification is required.

- (a) A Lipschitz function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is not C^1 and is not a contraction.
- (b) A sequence of functions $f_n : [0, 1] \rightarrow \mathbb{R}$ so that $|f_n(x)| \leq 1$ for all $x \in [0, 1]$ and all $n \geq 1$, but there is no uniformly convergent subsequence.

Question 2. Consider the set of points

$$S = \{(x, y) \in \mathbb{R}^2 : (x^2 + y^2)^2 - (x^2 - y^2) = 0\}.$$

Find all points (x_0, y_0) on the set S so that we cannot describe the set S in a neighborhood of (x_0, y_0) as the graph of a function of x (i.e., as $y = g(x)$).

Hint. Use a theorem to discard most points except a few candidates. Once you have a few candidates, you can see that S may not be a graph of a function around them based on the following two observations:

- There is no point on S with $|x| > 1$.
- Whenever $(x, y) \in S$, then the four points (x, y) , $(-x, y)$, $(x, -y)$, and $(-x, -y)$ are all in S . This implies that S must have a crossing at the origin.

Question 3. Suppose that (f_n) is an equicontinuous sequence in $C^0([a, b])$ and $p \in [a, b]$ is given.

- (a) If $(f_n(p))$ is a bounded sequence of real numbers, prove that (f_n) is uniformly bounded.
- (b) If we replace $[a, b]$ with an arbitrary metric space M , which of the following conditions would ensure that (a) is still true?
 - (i) M is bounded.
 - (ii) M is compact.
 - (iii) M is connected and compact.